

Day 3 – Dataset Expansion Attempt & Environment Failure Analysis

Date: 03/02/2026

Duration: ~4–5 hours

1. Objective of Day 3

The goal of Day 3 was to move beyond a basic proof-of-concept and begin **scaling the Ethiopian Sign Language (EthioSign) project** by:

- Expanding the dataset structure (starting with numbers 0–20)
 - Updating MediaPipe and dependencies to newer versions
 - Re-recording gestures with a cleaner, future-proof pipeline
 - Preparing the project for later extensions (speech output, larger vocabulary)
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2. What Was Attempted

Technical Tasks Attempted

- Create structured folders for numbers (0–20) under the dataset directory
- Re-run the data collection script using updated MediaPipe versions
- Fix repeated environment issues (cv2, mediapipe, venv problems)
- Adapt existing code to newer MediaPipe APIs

Conceptual Direction

- Shift the project from **word-only translation** toward a more complete system
 - Begin thinking about sign → text → speech scalability
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3. Code Intent (What the Code Was Supposed to Do)

The Day 3 code was intended to:

1. Open a camera feed (PC or phone camera)
2. Detect hands using MediaPipe Hands

3. Extract hand landmarks per frame
4. Save landmarks into CSV files per gesture class (numbers 0–20)
5. Organize data cleanly for future ML training

High-level pipeline (intended):

Camera → MediaPipe Hand Detection → Landmark Extraction → CSV Storage → Dataset

4. Why the Code Failed (Root Causes)

4.1 Environment Breakdown

- No consistent **virtual environment (venv)** was active
- Python packages were installed globally and conflicted
- `opencv-python (cv2)` was missing or improperly installed

4.2 MediaPipe Version Conflict

- Newer MediaPipe versions **removed or renamed APIs**
- Errors encountered included:
 - `mediapipe` has no attribute `solutions`
 - `mediapipe` has no attribute `image`
- Existing code relied on **older MediaPipe syntax**

4.3 Documentation Gap

- MediaPipe's newer releases lack clear beginner documentation
- Breaking changes were not backward-compatible
- Time was lost debugging instead of collecting data

4.4 Tooling Limitations

- Poor camera quality reduced landmark detection reliability
 - Phone camera integration (DroidCam) was delayed by slow internet
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5. Why Day 3 Is Officially Marked as a Failure

Day 3 is considered a failure because:

- No new usable dataset was collected
- Code never reached a stable, repeatable state

- Environment issues blocked progress entirely

However: this was a *technical failure*, not a conceptual one.

6. Key Lessons Learned

1. **Environment setup is non-negotiable**
 2. MediaPipe versioning must be locked early
 3. Scaling datasets without stable tooling is impossible
 4. Dependency failures are part of real AI engineering
 5. Documentation and planning matter as much as coding
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7. Strategic Decision Made

At the end of Day 3, the following decisions were made:

- Pause dataset expansion temporarily
 - Re-evaluate project scope and collaboration needs
 - Consider institutional collaboration (EAI, Deaf community)
 - Resume work later with a cleaner, locked-down environment
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8. Conclusion

Although Day 3 did not produce code or data, it played a **critical role** in identifying systemic weaknesses in the project setup. This failure directly informs a more robust strategy for future development.

Day 3 Outcome: ✗ Technical Failure

Day 3 Value: ✓ High learning impact